

DSA Phase 4 - Linked List, Recursion, and Backtracking (40 Problems)

Linked List

1. Reverse a Linked List
2. Detect Loop in Linked List
3. Remove Loop in Linked List
4. Find Middle of Linked List
5. Merge Two Sorted Linked Lists
6. Intersection Point in Y Shaped Linked Lists
7. Detect and Remove Loop in Linked List
8. Add 1 to a Number Represented as Linked List
9. Clone a Linked List with Random Pointers
10. Sort a Linked List

Recursion

1. Factorial of a Number using Recursion
2. Print N to 1 using Recursion
3. Sum of first N numbers
4. Check Palindrome using Recursion
5. Fibonacci Number using Recursion
6. Power Set of a String
7. Generate All Balanced Parentheses
8. Tower of Hanoi
9. Generate All Subsets of a Set
10. String to Integer Conversion using Recursion

Backtracking

1. N-Queens Problem
2. Sudoku Solver
3. Rat in a Maze
4. Word Search in 2D Grid
5. Subset Sum Problem

6. Palindrome Partitioning
7. Generate All Binary Strings Without Consecutive 1s
8. Combination Sum
9. Letter Combinations of a Phone Number
10. Permutations of a String